

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
Windows PowerShell
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PS C:\Users\dhars> wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [341 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [432 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.7 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [150 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [105 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [62.0 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [223 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [72.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [161 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Fetched 2292 kB in 3s (902 kB/s)
93 packages can be upgraded. Run 'apt list --upgradable' to see them.
dhars@DHARSHAN:/mnt/c/Users/dhars$ sudo apt install strace
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 93
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 3s (288 kB/s)
Selecting previously unselected package libunwind8:amd64.
```

```
(Reading database ... 42277 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd OSP
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP$ cd exec
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ ./program2
Cannot open source file: No such file or directory
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ echo "This is sample text for testing the file copy program." > sample.txt
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ nano sample.txt
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ ls -l
total 40
-rwxrwxrwx 1 dhars dhars 1292 Aug  5 04:06 program2.c
-rwxrwxrwx 1 dhars dhars 16472 Aug  5 03:22 program1
-rwxrwxrwx 1 dhars dhars 1056 Aug  5 03:22 program1.c
-rwxrwxrwx 1 dhars dhars 16216 Aug  5 04:06 program2
-rwxrwxrwx 1 dhars dhars 959 Aug  5 03:35 program2.c
-rwxrwxrwx 1 dhars dhars 55 Aug  5 04:10 sample.txt
This is sample text for testing the file copy program.
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ ./program2 sample.txt copy.txt
Cannot open source file: No such file or directory
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ strace -o trace_output.txt ./program2 sample.txt copy.txt
cat trace_output.txt
Cannot open source file: No such file or directory
execve("./program2", ["/.program2", "sample.txt", "copy.txt"], 0x7ffffd4ff6850 /* 26 vars */) = 0
brk(NULL) = 0x5fe66ef3b000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x748d34793000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20887, ...}) = 0
mmap(NULL, 20887, PROT_READ, MAP_PRIVATE, 3, 0) = 0x748d3478e000
close(3) = 0
```

```
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ strace -o cat_trace.txt cat sample.txt
cat cat_trace.txt
```

```

mmmap(0x73a60b628000, 1671168, PROT_READ|PROT_EXEC|MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x73a60b628000
mmmap(0x73a60b6c0000, 319488, PROT_READ|MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x73a60b6c0000
mmmap(0x73a60b80e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x73a60b80e000
mmmap(0x73a60b814000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x73a60b814000
close(3) = 0

```

